

Roman's guidance tests and first coronagraph images mark a preparation milestone

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A telescope may need to collect light for minutes or hours to reveal faint objects. During that time, even a tiny pointing error can blur an image. NASA's Nancy Grace Roman Space Telescope has completed tests showing that its guidance system can keep the observatory remarkably steady.

Engineers tested the fine-guidance system from September 15 to 21. Roman's main science instrument, the Wide Field Instrument, has 18 detectors. A small section of each detector rapidly monitors a different **guide star**: a star whose position is already known with great accuracy.

The spacecraft's attitude control system first points Roman toward the right region of sky. Fine guidance then tracks the reference stars and reports their positions about four times each second. Those measurements allow the control system to correct small drifts during an **exposure**.

The tests demonstrated stability better than one hundred-thousandth of a degree for half an hour during Wide Field Instrument observations, or eight hours during **Coronagraph** Instrument observations. The demonstrated steadiness is roughly comparable to keeping a laser aimed at a U.S. dime from about 240 kilometers away. Engineers expect further tuning to improve it.

Roman will also test a different approach to guidance. Instead of using only a star's pointlike image, it can use **spectra**, patterns showing how light is distributed across wavelengths. Roman already measures **spectra** for science, so the same information could help maintain its position. This guidance mode still needs **validation**.

The team tested fine guidance together with the **Coronagraph** Instrument on September 22 and 27. A **coronagraph** suppresses bright starlight, allowing much dimmer objects around stars to become visible. Roman's instrument is designed to demonstrate technologies for studying planets and dusty disks around nearby stars. It also has its own internal stability process.

In its first preliminary observation, the **coronagraph** imaged a faint star in the Large Magellanic Cloud, a neighboring galaxy. The detectors were deliberately warmer than their eventual operating temperature to help prevent contamination from sticking to them. That produced extra image noise.

For the second observation, the team cooled the detectors to improve **sensitivity** and pointed toward a location expected to contain many stars. Multiple stars appeared, confirming the pointing.

These observations checked that light could pass through the instrument and form a focused image. They begin a careful sequence of increasingly demanding tests.

Word Treasure

guide star -- A star with an accurately known position used as a pointing reference.

exposure -- The time during which an instrument collects light for an image.

spectra -- Patterns showing how light is distributed across different wavelengths.

validation -- Testing to check whether something works as intended.

coronagraph -- An instrument that suppresses bright starlight to reveal fainter objects.

sensitivity -- An instrument's ability to detect faint signals, such as dim light.

Background you need

Guide stars provide known reference positions that help a telescope correct small drifts. Suppressing bright starlight helps scientists study much dimmer objects around stars. Spectral guidance would use information Roman already collects for science.

Break it down

Explains why long observations require steady pointing.
Describes the detectors, guide stars and control system.
Reports measured stability and introduces spectral guidance awaiting validation.
Explains the coronagraph's purpose and additional stability process.
Describes two preliminary observations and the tests still ahead.

Why it matters

The tests check the pointing and imaging abilities Roman needs before attempting more demanding observations of faint objects.

Test what you remember

1. How many detectors does Roman's Wide Field Instrument have?
 - (A) 27 detectors
 - (B) 18 detectors
 - (C) 4 detectors
 - (D) 8 detectors
2. What does the attitude control system do first?
 - (A) Measures spectra to validate spectral guidance
 - (B) Points Roman toward the right sky region
 - (C) Cools Roman's detectors for greater sensitivity
 - (D) Suppresses starlight to reveal faint planets
3. How long did the tests demonstrate the stated stability for Coronagraph Instrument observations?
 - (A) Half an hour
 - (B) One minute
 - (C) Eight hours
 - (D) Four minutes
4. What information would Roman's spectral guidance use?
 - (A) Temperature patterns across different detectors
 - (B) Movement patterns across neighboring galaxies
 - (C) Dust patterns around nearby planets
 - (D) Light patterns across different wavelengths
5. Why did the first preliminary image contain extra noise?
 - (A) The detectors were deliberately kept warmer
 - (B) The coronagraph had completed all testing
 - (C) The telescope stopped collecting incoming light
 - (D) The guide stars had unknown positions
6. What did the appearance of multiple stars in the second observation confirm?
 - (A) Roman had finished validating its spectral guidance
 - (B) Roman was pointing toward the expected location
 - (C) Roman was using warmer detectors for sensitivity
 - (D) Roman had completed its future planet studies

Answer key (try the quiz first!): 1: B 2: B 3: C 4: D 5: A 6: B

MY THOUGHTS (at least 20 words):